



Semester II Examinations 2017/2018

Exam Code(s)	3BME1, 3BMS2, 3BS9, 4BMS2, 4BS2, 4FM2
Exam	3rd Arts, 3rd Science, 4th Arts, 4th Science
Module	Numerical Analysis II
Module Code	MA378
External Examiner(s)	Professor T. Brady
Internal Examiner(s)	Professor G. Ellis Dr N. Snigireva*
Instructions	Answer THREE questions. All questions carry equal marks.
Duration	2 hours
No. of Pages	4 pages (including this cover page)
Discipline	Mathematics
Requirements:	
Release to Library	Yes
Release in Exam Venue	Yes
MCQ	No
Statistical Tables/ Log Tables	Yes

1. Let $f(x) = x^2 e^x$ and let $x_0 = -1$, $x_1 = 0$ and $x_2 = 1$.

(a) Construct polynomials $l_0(x)$, $h_1(x)$, $h_2(x)$, $k_1(x)$ and $k_2(x)$ of degree 4 such that the polynomial

$$p_4(x) = f(x_0)l_0(x) + \sum_{i=1}^2 (f(x_i)h_i(x) + f'(x_i)k_i(x))$$

satisfies the conditions

$$p_4(x_i) = f(x_i), \quad i = 0, 1, 2$$

and

$$p'_4(x_i) = f'(x_i), \quad i = 1, 2.$$

[16 marks]

(b) Show that for each x in $[-1, 1]$ there is a number $\tau \in (-1, 1)$, depending on x , such that

$$f(x) - p_4(x) = \frac{(x+1)x^2(x-1)^2}{120} f^{(5)}(\tau),$$

where $f^{(5)}$ denotes the 5th derivative of f . **[14 marks]**

2. Suppose that the interpolation points $x_0 < x_1 < \dots < x_n$ are equally spaced and $x_0 = 0$ and $x_n = 1$. Let S denote the natural cubic spline interpolant to a function f , and s_i its restriction to $[x_{i-1}, x_i]$.

(a) If we let $\sigma_i = S''(x_i)$ for $i = 0, \dots, n$ show that

$$s_i(x) = \alpha_i(x - x_{i-1}) + \beta_i(x_i - x) + \frac{n\sigma_{i-1}}{6}(x_i - x)^3 + \frac{n\sigma_i}{6}(x - x_{i-1})^3,$$

where

$$\alpha_i = nf_i - \frac{\sigma_i}{6n}, \quad \text{and} \quad \beta_i = nf_{i-1} - \frac{\sigma_{i-1}}{6n}.$$

[continued on the next page]

Hence deduce that S can be determined by solving the system

$$\begin{aligned}\sigma_0 &= 0 \\ \frac{1}{6}(\sigma_{i-1} + 4\sigma_i + \sigma_{i+1}) &= n^2(f_{i-1} - 2f_i + f_{i+1}) \quad \text{for } i = 1, 2, \dots, n-1 \\ \sigma_n &= 0.\end{aligned}\tag{1}$$

[14 marks]

(b) Let $f(x) = x^3$.

(i) Show that $f(x) \neq S(x)$. **[6 marks]**

(ii) Show that if we replace $\sigma_0 = 0$ and $\sigma_n = 0$ by $\sigma_0 = f''(0)$ and $\sigma_n = f''(1)$ in system (1) then $f(x) = S(x)$. (You can use without a proof the fact that system (1) of linear equations has unique solution.) **[10 marks]**

3. (a) (i) State Simpson's rule and show that it has precision 3.

[10 marks]

(ii) Let w_0, w_1 and w_2 be quadrature weights and $x_1 \in [-1, 1]$. Use the method of undetermined coefficients to show that the 3-point rule

$$\int_{-1}^1 f(x) dx \approx w_0 f(-1) + w_1 f(x_1) + w_2 f(1).$$

coincides with Simpson's rule on $[-1, 1]$. **[5 marks]**

(b) Let $\{\tilde{p}_i\}_{i=0}^\infty$ be the sequence of orthogonal monic polynomials with respect to the inner product

$$(f, g) := \int_a^b f(x)g(x) dx$$

(i) Prove that $\tilde{p}_i$ is orthogonal to all polynomials of degree less than i . **[5 marks]**

[continued on the next page]

- (ii) Use Part (i) to show that the sequence $\{\tilde{p}_i\}_{i=0}^{\infty}$ can be constructed as follows. Let α_{n+1} and β_{n+1} be defined as

$$\alpha_{n+1} = \frac{(x\tilde{p}_n, \tilde{p}_n)}{(\tilde{p}_n, \tilde{p}_n)} \quad \text{and} \quad \beta_{n+1} = \frac{(x\tilde{p}_n, \tilde{p}_{n-1})}{(\tilde{p}_{n-1}, \tilde{p}_{n-1})},$$

then the sequence is given by

$$\tilde{p}_0(x) \equiv 1, \quad \tilde{p}_1(x) = x - \alpha_1$$

and

$$\tilde{p}_{n+1}(x) = (x - \alpha_{n+1})\tilde{p}_n(x) - \beta_{n+1}\tilde{p}_{n-1}(x),$$

for $n \geq 1$. **[10 marks]**

4. (a) (i) Let $x_0, \dots, x_n$ be the zeros of $\tilde{p}_{n+1}$, the $(n+1)$ th polynomial in the sequence of orthogonal monic polynomials $\{\tilde{p}_i\}_{i=0}^{\infty}$ with respect to the inner product $(f, g) := \int_a^b f(x)g(x) dx$. Set

$$G_n(f) = w_0 f(x_0) + \dots + w_n f(x_n)$$

where $\{w_i\}_{i=0}^n$ are quadrature weights. Show that $G_n(f)$ has precision $2n+1$. **[12 marks]**

- (ii) Show that it is impossible to choose $n+1$ quadrature points and weights so that the $n+1$ -point quadrature rule

$$\int_a^b f(x) dx \approx \sum_{k=0}^n w_k f(x_k)$$

has precision $2n+2$. **[6 marks]**

- (b) Consider the following differential equation:

$$-u''(x) = \exp(x) \text{ on } (0, 1), \quad \text{and} \quad u(0) = u(1) = 0. \quad (2)$$

- (i) State the variational formulation of the differential equation in (2) and show that the solution to the variational problem is unique. **[8 marks]**
- (ii) Suppose one wanted to compute an approximation to (2) using the finite element method (FEM) with the usual piecewise linear Galerkin basis functions on the uniform mesh $\{x_0, x_1, \dots, x_n\}$. Give a formula for the basis functions. How is the FEM solution expressed in terms of this basis? **[4 marks]**